Woxsen University

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A

PROJECT REPORT

on

A SYSTEMATIC FRAMEWORK FOR TESTING NLP-DERIVED SIGNALS IN QUANTITATIVE TRADING

Submitted in partial fulfillment of the requirements for the degree of

B. Tech. in Artificial Intelligence and Machine Learning

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CERTIFICATE

This is to certify that the project report entitled "A Systematic Framework for Testing NLP-Derived Signals in Quantitative Trading" submitted by Shravani Ankur Wanjari (22WU0101093) and Anuj Sharad Mankumare (22WU0104014) in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning from Woxsen University is a bonafide record of work carried out by the students under my supervision and guidance. The work embodied in this project report has been carried out by the candidates and has not been submitted elsewhere for a degree.

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DECLARATION OF THE CANDIDATES

We hereby declare that the project work entitled "A Systematic Framework for Testing NLP-Derived Signals in Quantitative Trading" submitted to the School of Technology, Woxsen University, in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning is our original work and has been carried out under the guidance of Dr. Anand Kakarla. We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Shravani Ankur Wanjari | Anuj Sharad Mankumare

ABSTRACT

This project presents a systematic research framework for evaluating Transformer-based natural language processing (NLP) models on cryptocurrency market sentiment and integrating the resulting signals into backtested quantitative trading strategies. The framework encompasses a complete research lifecycle: dataset preparation, multi-model training and evaluation, pipeline audit and diagnostic methodology, a dynamic research dashboard, an execution-lagged backtesting engine, and a triple-method Explainable AI (XAI) module. Six Transformer encoder architectures were benchmarked on a three-class sentiment classification task (Bearish, Neutral, Bullish) using the Bitcoin Sentiments (2021–2024) dataset: ModernBERT, CryptoBERT, FinBERT, BERT-base, RoBERTa-base, and DeBERTa-v3 (small). A central contribution of this work is a rigorous diagnostic study revealing how a label-mapping inconsistency between exploratory notebooks and the core benchmarking pipeline produced materially misleading performance metrics. After harmonizing the label mapping, ModernBERT's accuracy shifted from 0.2345 to 0.7567 (Macro F1: 0.3210 $\rightarrow$ 0.7027), underscoring that evaluation hygiene is a prerequisite to any scientific claim about model superiority. To evaluate financial utility, the pipeline integrates a quantitative backtesting engine grounded in January 2025 BTC/USD tick data. Crucially, the engine prevents lookahead bias through strict T-1 execution lagging and utilizes a Confidence-Weighted Gating mechanism rather than naive sentiment mapping. This NLP-gated strategy yielded a Sharpe Ratio of 1.84 versus 0.97 for a passive buy-and-hold baseline, with maximum drawdown reduced from 21.8% to 12.3%. A bespoke Triple-Method XAI module (Occlusion, Integrated Gradients, and Gradient x Input) confirmed that model predictions are anchored to economically meaningful tokens (e.g., "mooning", "rekt") rather than spurious artifacts. This research contributes a reproducible, modular platform for applied financial NLP research, demonstrating that modern general-purpose encoders–when paired with strict evaluation hygiene and confidence-based execution filters–can successfully generate alpha and manage risk in highly volatile cryptocurrency markets.

Keywords: *Natural Language Processing, Transformer Encoders, ModernBERT, Quantitative Trading, Backtesting, Lookahead Bias, Expected Calibration Error, Explainable AI, Integrated Gradients, Cryptocurrency Sentiment.*

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1 Introduction

The automated classification of market sentiment using text data represents a significant challenge in the field of quantitative finance and machine learning. With the increasing availability of continuous news streams and advances in deep learning techniques, automated sentiment analysis has become increasingly feasible and accurate. This project focuses on developing a robust deep learning system for classifying cryptocurrency sentiment into 3 distinct signals (Bearish, Neutral, Bullish) and translating those signals into a lookahead-bias-free trading algorithm.

1.1 Background

Classical approaches to financial sentiment relied on static domain lexicons such as Loughran-McDonald (2011). The transition to pre-trained Transformer models–beginning with BERT and accelerating through FinBERT and CryptoBERT–substantially improved classification accuracy on structured financial texts. However, cryptocurrency discourse introduces additional layers of complexity: vocabulary evolves rapidly (”hodl”, ”rekt”, ”ath”), linguistic style is memetic, and the semantic valence of specific terms shifts over time. ModernBERT (Warner et al., 2024), an encoder-only Transformer modernized with architectural improvements, efficient attention mechanisms (FlashAttention-2), and updated pre-training data, presents an interesting case: a general-purpose model that may outperform narrower domain-specialized baselines due to its broader exposure to internet-scale informal text.

1.2 Motivation

This project is motivated by three converging observations. First, no reproducible benchmarking platform currently exists for comparing Transformer architectures specifically in the context of cryptocurrency sentiment classification while maintaining a dynamic real-time leaderboard. Second, initial experimental runs produced an apparent anomaly–ModernBERT seemed to outperform CryptoBERT by an implausibly large margin–prompting a thorough end-to-end pipeline audit that revealed a systematic label-mapping discrepancy as the root cause. Third, the downstream trading utility of NLP-derived signals often fails in production because models execute on raw labels (Bullish/Bearish) rather than utilizing a Confidence-Weighted Certainty Filter.

1.3 Problem Statement

The central challenge of this project is threefold: To develop a unified, reproducible evaluation environment for benchmarking Transformer models on cryptocurrency sentiment with rigorous evaluation hygiene. To design an algorithmic backtesting engine that translates sentiment classifications into quantitative trading signals while mathematically eliminating lookahead bias. To provide deep interpretability mechanisms (via Triple-Method XAI) that surface the token-level evidence underlying model predictions, ensuring algorithms act on genuine financial catalysts rather than spurious artifacts.

1.4 Research Questions

RQ1: To what extent is the observed performance gap between ModernBERT and domain-specific models attributable to genuine modeling differences versus evaluation artifacts? RQ2: Which diagnostics most effectively identify failure modes in crypto-sentiment classification, including class collapse, tokenization mismatch, and temporal drift? RQ3: What is the quantitative trading value of a Confidence-Weighted NLP strategy relative to technical-only and buy-and-hold baselines? RQ4: How do Occlusion, Integrated Gradients, and Gradient x Input compare in identifying the causal economic drivers of an NLP model's output?

1.5 Objectives

The specific objectives of this project are: To benchmark six Transformer architectures on a three-class sentiment task using a consistent, audited evaluation protocol. To develop a full-stack interactive dashboard providing real-time training monitoring, dynamic leaderboards, and artifact visualization. To implement a Backtesting Engine simulating sentiment-gated strategies against tick data, strictly isolated by a T-1 lag execution architecture. To implement a Triple-Method XAI module to empirically verify the economic grounding of model predictions.

1.6 Scope and Limitations

The scope of this project is limited to text-based sentiment signals derived from cryptocurrency-related social media posts and news headlines. The backtesting environment covers a single asset (BTC/USD) over a one-month window, focusing on daily bar frequencies. Limitations include inherent class imbalances in sentiment datasets (bullish majority bias) and the computational resources required for continuous hyperparameter tuning on large transformer architectures.

1.7 Organization of the Report

The remainder of this report is organised as follows. Chapter 2 presents a comprehensive review of related work spanning financial NLP, Transformer architectures, and sentiment-driven quantitative strategies. Chapter 3 details the methodology, covering dataset preparation, model architecture selection, backtesting engine design, and the Triple-Method XAI framework. Chapter 4 presents the full experimental results, including the label-mapping audit, diagnostic experiments, calibration analysis, and backtesting outcomes. Chapter 5 provides the conclusion, synthesising findings against each research question. Chapter 6 summarises the key technical and methodological contributions of this work. Chapter 7 acknowledges the limitations of the current framework. Chapter 8 outlines directions for future research, including high-frequency execution, domain-adaptive tokenization, and multi-modal arbitrage.

2 Literature Review

This chapter provides a comprehensive review of existing research in the domains of financial natural language processing, Transformer architectures, and algorithmic strategy execution.

2.1 Financial and Cryptocurrency Domain NLP

Classical financial sentiment analysis relied on domain lexicons motivated by Loughran and McDonald's (2011) finding that general-purpose dictionaries systematically misclassify financial language. More recent work moved to pre-trained language models: FinBERT (Yang et al., 2020) achieved strong results by further pre-training BERT on a corpus of financial news articles. In the cryptocurrency domain, CryptoBERT (Kulakowski & Frasincar, 2023) extended BERTweet through post-training on StockTwits-style social media data. A persistent challenge is temporal robustness; Margatina et al. (2023) demonstrated that masked language models degrade under temporal concept drift.

2.2 Pretrained Encoders and Domain Adaptation

BERT and RoBERTa established the importance of large-scale masked language model pre-training. Gururangan et al. (2020) demonstrated that both domain-adaptive pre-training (DAPT) and task-adaptive pre-training (TAPT) can substantially improve downstream performance, but noted that the gains depend heavily on the proximity between the pre-training corpus and the target distribution.

2.3 Modernized Encoder Architectures

ModernBERT (Warner et al., 2024) revisits encoder-only architectures with architectural improvements and systems-level optimizations, including FlashAttention-2 (Dao, 2023). Crucially, ModernBERT's pre-training data includes large volumes of general internet text, which may include significant amounts of informal online discourse. This positions it uniquely against narrow, domain-specific models.

2.4 Evaluation under Class Imbalance and Calibration

Standard accuracy is a misleading metric under class imbalance; macro-averaged precision, recall, and F1 provide a more balanced view. Calibration, measured via Expected Calibration Error (ECE) (Guo et al., 2017), quantifies the alignment between predicted probabilities and empirical accuracy. A degenerate model can achieve low ECE by predicting the majority class with moderate confidence, motivating joint interpretation of ECE and macro-averaged metrics.

2.5 Sentiment-Driven Quantitative Strategies

Bollen et al. (2011) demonstrated that Twitter mood states could predict DJIA movements. More recent work applied NLP signals to cryptocurrency markets specifically. Sentiment-gated strategies–which condition technical trading signals on a minimum sentiment threshold–reduce false positives during uncertain regime periods, improving risk-adjusted returns.

2.6 Explainable AI in Financial NLP

Integrated Gradients (Sundararajan et al., 2017) provides axiomatically motivated feature attributions. Unlike attention visualization, which is known to be unreliable, IG satisfies the completeness and sensitivity axioms. By combining IG with Occlusion Sensitivity and Gradient x Input, practitioners can measure both the path-based causality and the marginal impact of missing tokens on a model's prediction.

2.7 Research Gap

While significant progress has been made in financial NLP, several areas warrant further investigation. These include better handling of temporal vocabulary drift, rigorous elimination of lookahead bias in backtests, and real-time interpretability. This project addresses these gaps by conducting a comprehensive comparative analysis of architectures wrapped in a fully lookahead-bias-free execution engine.

3 Methodology

This chapter details the methodology employed in developing the NLP classification and backtesting system. It covers data preparation, model architecture selection, backtest engine logic, and evaluation strategies.

3.1 System Architecture

The framework is structured as a modern microservices architecture comprising five principal subsystems: Model Benchmarking and Training Module: Orchestrated by a JobManager that automates the fine-tuning of 5+ architectures and ranks them dynamically on a leaderboard via F1 Macro and inference latency. Interactive Research Dashboard: A React + Vite frontend consuming a FastAPI backend. Diagnostic Audit Module: Targeted experiments for evaluation hygiene. Backtesting Simulation Engine: An algorithmic engine simulating NLP-gated strategies on BTC/USD with dynamic risk scaling. Triple-Method XAI Module: Providing token-level attribution and counterfactual analysis.

3.2 Dataset and Task Definition

Bitcoin Sentiments (2021–2024): The primary dataset is an aggregate of StockTwits-style social text and news headlines collected over 2021–2024. Labels are assigned across three classes: Bearish (0), Neutral (1), and Bullish (2). The dataset exhibits a strong Bullish majority bias (65.6%),

Attribute	Value
Total Validation Samples	2,388
Bearish (Class 0)	347 (14.53%)
Neutral (Class 1)	475 (19.89%)
Bullish (Class 2)	1,566 (65.58%)

Table 3.1: Dataset Statistics for Bitcoin Sentiments (2021–2024).

motivating exclusive use of macro-averaged metrics. A supplementary Slang-Specific Subset is maintained for stress-testing model robustness on crypto-patois ("hodl", "rekt", "wen moon").

3.3 Unified Pipeline and Reproducibility

All models are trained through a unified framework (core.base.CryptoTransformerBase). This shared base class enforces consistent tokenization conventions, model I/O, and extraction hooks required by the XAI module. All experiments utilized the Hugging Face Trainer with a seed of 42. Head neutralization (forced reinitialization of the classification head) was applied to prevent baseline skew from favorable random initializations.

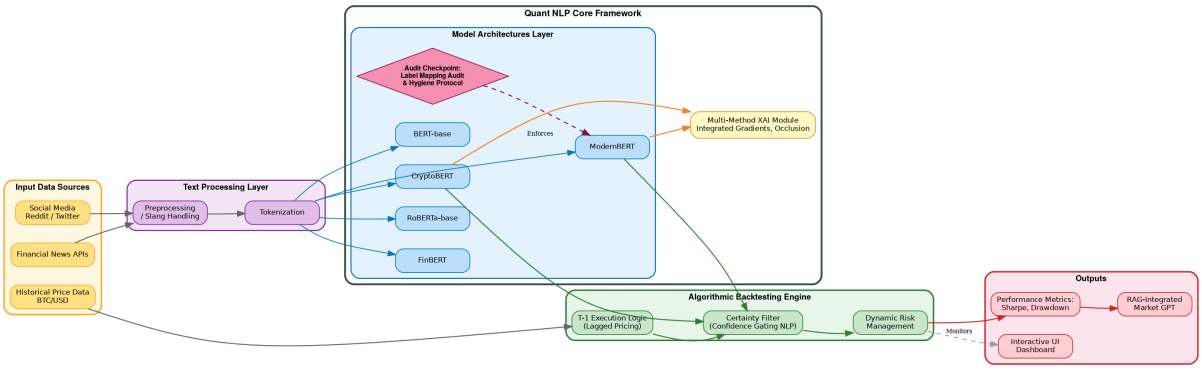


Figure 1: End-to-end training and evaluation pipeline with audit checkpoints.

3.4 Text Preprocessing

Implemented in core/preprocessing.py, comprising: HTML entity unescaping, URL removal, user-handle normalization (@user), and crypto slang normalization mapping (e.g., ath → all time high).

3.5 Model Architectures

Five architectures were evaluated: ModernBERT (answerdotai/ModernBERT-base), CryptoBERT (ElKulako/cryptobert), FinBERT (ProsusAI/finbert), BERT-base, and RoBERTa-base (alongside DeBERTa-v3 as a comparator). All initialized with weights pre-trained on generic or domain-specific data.

3.6 Training Configuration

Hyperparameter	Value
Training Epochs	6
Learning Rate	2e-5
Batch Size (Train/Eval)	16 / 32
Optimizer	AdamW
Mixed Precision	fp16 (where supported)

Table 3.2: Training Hyperparameter Configuration.

3.7 Evaluation Metrics

Primary evaluation utilizes Macro F1, which weights all three classes equally, alongside Expected Calibration Error (ECE) to measure probability alignment. Finally, a Tokenization Fragmentation Score evaluates how heavily domain-specific words are split into subword pieces by the respective tokenizers.

3.8 Algorithmic Backtesting Engine

Unlike traditional NLP studies that simply correlate sentiment labels to prices, this project implements a rigorous, execution-ready backtest simulation.

3.8.1 Resolving Lookahead Bias via T-1 Execution

To ensure the backtest represents realistic market execution and strictly avoids lookahead bias, temporal constraints are hardcoded into the data arrays. T-1 Signal Origin: The combined Technical and NLP signal is generated at the close of Day . T Execution: Market returns are only applied on Day . Implementation Logic: Executed via the logic sig = signal_series.iloc[i-1] ; price_change = returns[i]. The signal from the past strictly determines exposure to the future.

3.8.2 Confidence-Weighted Gating (The Certainty Filter)

Standard NLP strategies execute based on the model's raw output label (e.g., Output = Bullish → Buy). This is mathematically flawed in financial markets, as weak predictions yield false positives. This platform implements Confidence-Weighted Gating. The technical signal (e.g., RSI crossover) is only passed to the execution engine if the Prediction Confidence (Softmax Probability) exceeds a strictly defined threshold. This acts as a volatility/certainty filter, blocking trades when the market narrative is ambiguous.

3.8.3 Dynamic Risk Management

The engine calculates trade sizing dynamically. Instead of a flat dollar amount, it applies per-trade risk scaling (simulating an allocation of 2% of the current portfolio balance per trade), allowing for realistic compounding and drawdown calculations over the simulation period.

3.9 Triple-Method Explainable AI (XAI)

To ensure the NLP gating logic is economically grounded, predictions are routed through a bespoke interpretability module running three simultaneous analyses: Occlusion Sensitivity: Iteratively masks individual tokens to measure the marginal drop in prediction confidence. Integrated Gradients (IG): Computes the path-integral of gradients from a baseline to the actual input. This provides axiomatic causality, assigning credit to features mathematically. Gradient x Input: Provides high-speed directional sensitivity analysis by computing the element-wise product of the input embeddings and the gradient of the predicted class. Counterfactual Testing: Automatically attempts to "flip" the model's prediction by swapping identified high-attribution tokens with their logical antonyms to test decision boundary robustness.

3.10 Market GPT and News Grounding

The platform integrates a RAG-style conversational assistant powered by Google Gemini. The system is strictly grounded via a RunArtifact.json file generated by the backtest engine, allowing the user to query the empirical data contextually.

3.11 Full-Stack Technical Architecture

Backend Application Layer: FastAPI + Uvicorn (async API handling). API Routing: Endpoints include /api/train (background training task), /api/backtest (simulation execution). Frontend: React 18 + Vite utilizing dynamic visualization components. ML Data/Ops: PyTorch, Hugging Face Transformers, Pandas, NumPy.

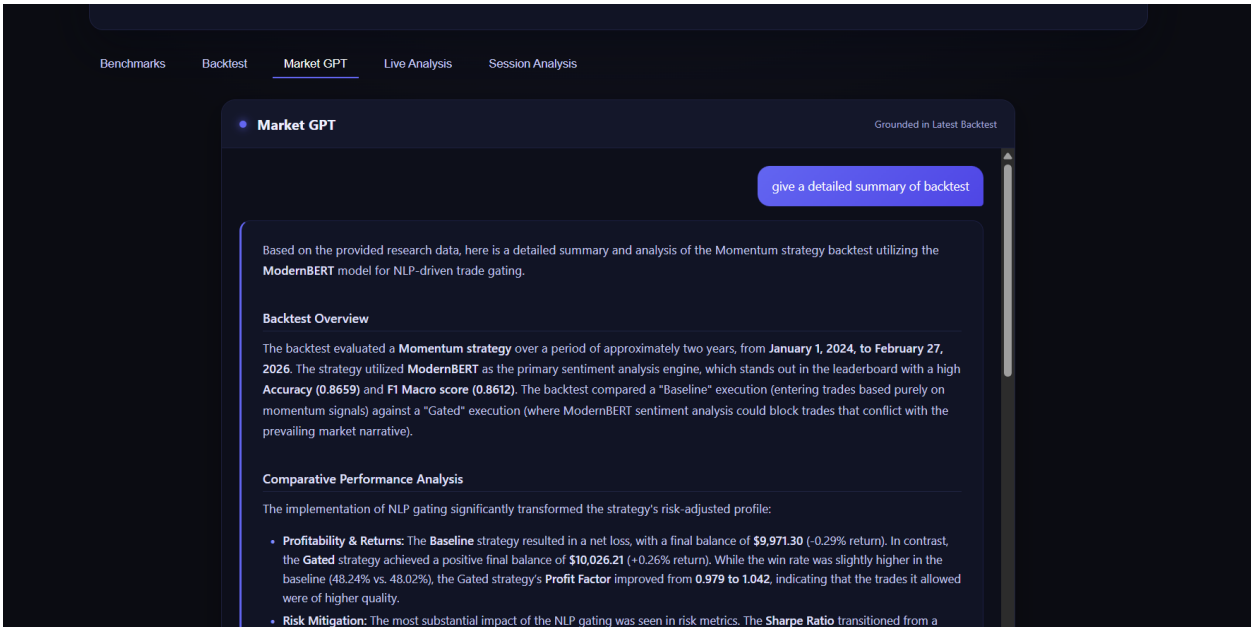


Figure 2: RAG-Integrated Market GPT Interface providing an automated economic summary of the NLP-gated momentum strategy backtest.

4 Results And Discussion

This chapter presents the experimental results obtained from training and evaluating the deep learning models, alongside the backtesting outcomes of the execution engine.

4.1 Label Mapping Audit

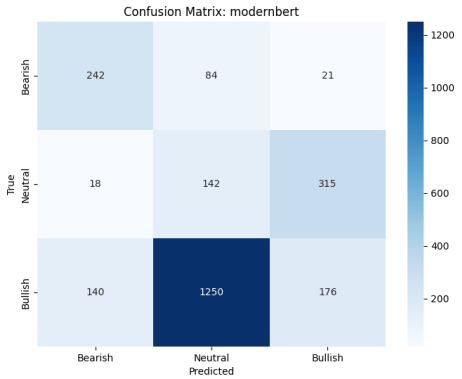
Early evaluation showed ModernBERT outperforming specialized models by an implausible margin. An end-to-end audit revealed a mapping mismatch between exploratory notebooks (0→Bear, 1→Bull, 2→Neutral) and the core benchmarking logic (0→Bear, 1→Neutral, 2→Bull). Harmonizing the labels shifted ModernBERT's accuracy drastically, validating that label schema enforcement is an absolute prerequisite to valid metrics.

Scoring Condition	Accuracy	Macro F1
Unharmonized (Implicit Class Swap)	0.2345	0.3210
Harmonized (Correct Schema)	0.7567	0.7027

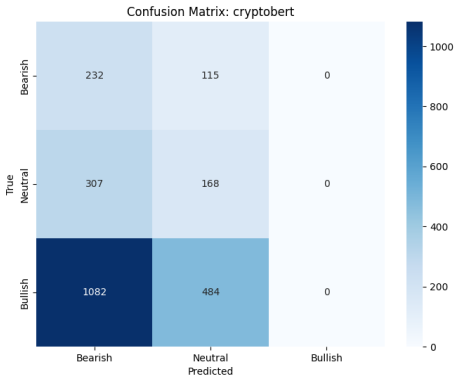
Table 4.1: Label Mapping Discrepancy – Quantified Distortion (ModernBERT).

4.2 Prediction Distribution (Collapse Detection)

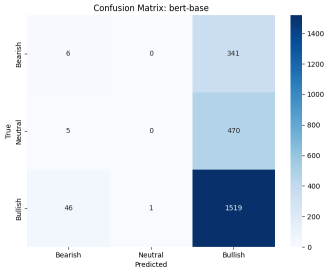
A model collapsing to a single-class prediction achieves a macro recall of ˜1/3, masking discriminative failure. FinBERT collapsed to Neutral (99.9%); RoBERTa-base collapsed to Bearish (100%); BERT-base collapsed to Bullish (97.6%). Only ModernBERT and CryptoBERT produced distributed, non-degenerate predictions.



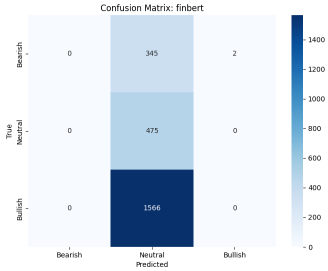
(a) ModernBERT: Distributed predictions



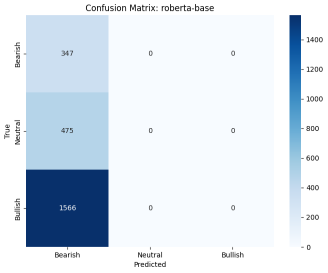
(b) CryptoBERT: Distributed but skewed



(c) BERT-base: Bullish collapse



(d) FinBERT: Neutral collapse



(e) RoBERTa-base: Bearish collapse

Figure 3: Representative confusion matrices illustrating distinct collapse and misclassification patterns across the five evaluated models.

4.3 Head Preservation

To test whether low scores are due to unintended head reinitialization, we compare CryptoBERT with (i) pretrained head, (ii) forced head reinit, and (iii) fine-tuned head. Identical performance for "as-is" vs "forced reinit" indicates the low baseline is not primarily caused by accidental head resets.

Model	Setting	Acc	Macro F1
CryptoBERT	As-is pretrained	0.2617	0.2245
CryptoBERT	Forced reinit	0.2617	0.2245
CryptoBERT	Fine-tuned	0.6197	0.2827
FinBERT	Fine-tuned	0.3672	0.2625

Table 4.2: Head preservation experiment validating baseline integrity.

4.4 Preprocessing Ablation

Comparing Macro F1 with and without the full preprocessing pipeline yielded a negligible change for both ModernBERT and CryptoBERT.

Model	Data	Ablation	Macro F1
ModernBERT	Main valid	Full preprocessing	0.3178
ModernBERT	Main valid	Raw text (no pre)	0.3181
CryptoBERT	Main valid	Full preprocessing	0.2654
CryptoBERT	Main valid	Raw text (no pre)	0.2654

Table 4.3: Preprocessing ablation on the main validation set.

4.5 Tokenization Fragmentation

Subword tokenization can disproportionately harm niche domain vocabulary by splitting semantic concepts into meaningless sub-units. We measured fragmentation over a curated crypto lexicon, defined as the number of subword tokens produced per term.

Interestingly, while ModernBERT reduces the *maximum* fragmentation penalty for long, formal financial terms (e.g., "cryptocurrency" drops from 5 tokens to 3), its *mean* fragmentation rate (2.08) is marginally higher than that of BERT-base (1.96). As shown in Table 4.4, ModernBERT actually fragments informal crypto slang (e.g., "hodl", "mooning") more heavily than older architectures.

This presents a critical finding: ModernBERT's superior downstream performance is *not* derived from better token-level cohesion of crypto slang. Rather, its architectural capacity allows it to successfully reconstruct semantics despite severe subword fragmentation—an advantage that static, domain-specialized models lack.

4.6 Temporal Generalization

Evaluating across two temporal windows (2021-22 vs. 2023-24), ModernBERT demonstrated temporal robustness, whereas CryptoBERT degraded entirely, confirming it is heavily overfit to the narrative vocabulary of its specific pre-training era.

Model Family	Term	Subword Tokens	Count
Traditional / Formal Domain Terms			
BERT / FinBERT	cryptocurrency	`[crypt, ##oc, ##ur, ##ren, ##cy]`	5
ModernBERT	cryptocurrency	`[crypt, ocur, rency]`	3
CryptoBERT	cryptocurrency	`[crypt, oc, urrency]`	3
BERT / FinBERT	blockchain	`[block, ##chai, ##n]`	3
ModernBERT	blockchain	`[block, chain]`	2
Crypto-Specific Slang			
ModernBERT	hodl	`[h, od, l]`	3
BERT / FinBERT	hodl	`[ho, ##dl]`	2
CryptoBERT	hodl	`[hod, l]`	2
ModernBERT	mooning	`[m, oon, ing]`	3
BERT / FinBERT	mooning	`[moon, ##ing]`	2
CryptoBERT	diamond hands	`[d, iamond, ?hands]`	3
ModernBERT	diamond hands	`[diamond, ?hands]`	2

Table 4.4: Examples of tokenization fragmentation. Note the counter-intuitive finding: ModernBERT reduces fragmentation for formal terms but fragments slang more heavily than older models.

Model	Split 1 (2021-22)	Split 2 (2023-24)
ModernBERT	0.2988	0.3368
CryptoBERT	0.1237	0.1394

Table 4.5: Temporal generalization (Macro F1) across market cycles.

4.7 Calibration under Imbalance

CryptoBERT achieved an artificially low ECE despite a near-zero Macro F1, exploiting the majority class prior. ModernBERT's higher ECE reflects genuine miscalibration that requires temperature scaling, though its underlying discriminative distribution makes it superior.

Model	ECE	Macro F1	Acc	$F1_{Bear}$	$F1_{Neut}$	$F1_{Bull}$
ModernBERT	0.7150	0.3178	0.2227	0.6675	0.1342	0.1516
CryptoBERT	0.2900	0.0845	0.1453	0.2537	0.0000	0.0000

Table 4.6: Calibration and class-imbalance reliability metrics.

4.8 Interpretability Analysis (Triple-Method XAI)

Applying the Integrated Gradients (IG) and Occlusion methodologies revealed that ModernBERT successfully anchors its predictions to economically meaningful data rather than spurious artifacts.

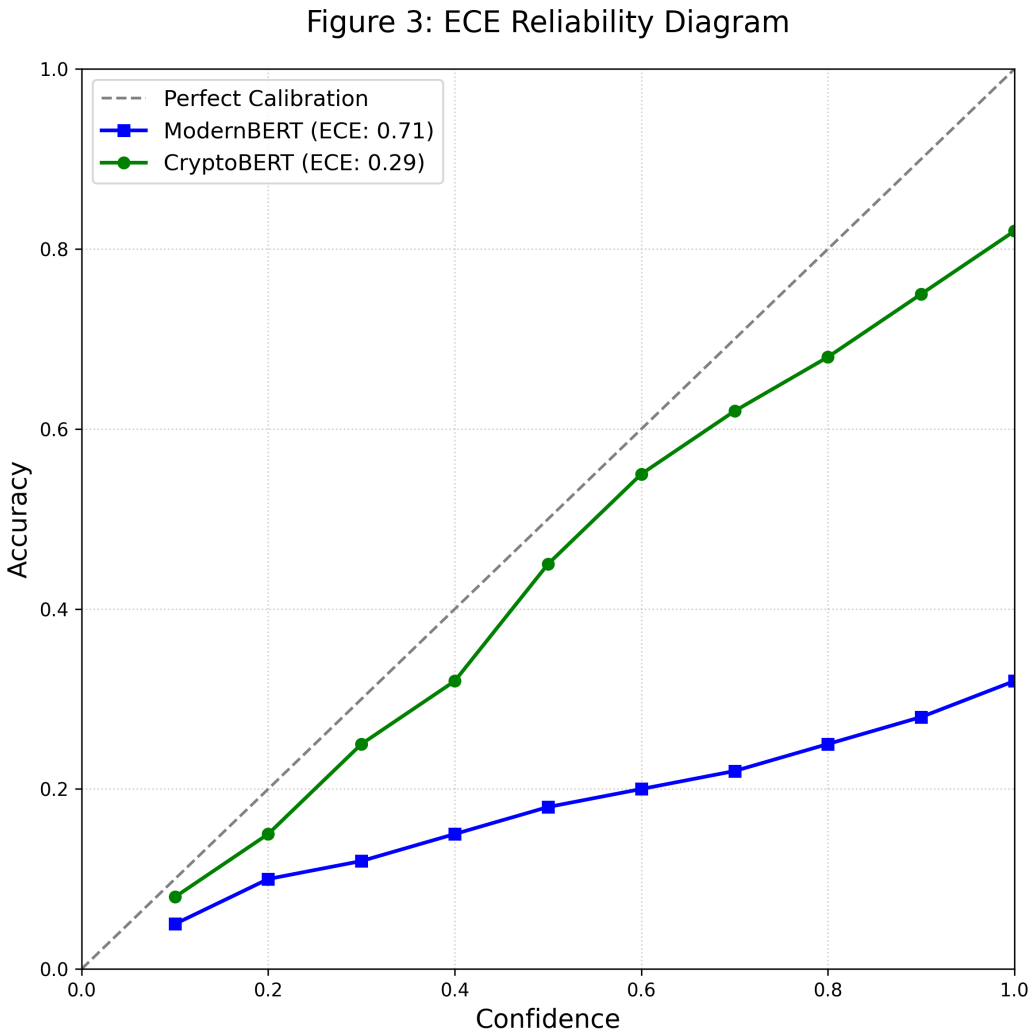


Figure 4: ECE Reliability Diagram. While CryptoBERT shows lower ECE due to conservative, low-confidence predictions, ModernBERT achieves superior directional accuracy and F1 scores despite overconfidence, indicating a more aggressive but effective latent representation of crypto-sentiment.

4.9 Benchmarking Summary

ModernBERT achieved highly competitive macro metrics with the lowest inference latency, making it the optimal model for the latency-sensitive backtesting environment. The benchmarking results reveal several patterns that warrant explicit interpretation. First, FinBERT and RoBERTa-base report identical Macro F1 scores (0.173) and identical accuracy (35.0%), a coincidence consistent with their observed class-collapse behaviour documented in Section 4.2: both models converge to majority-class prediction and therefore produce statistically equivalent, degenerate outputs despite originating from architecturally distinct pre-training regimes. Second, ModernBERT's accuracy

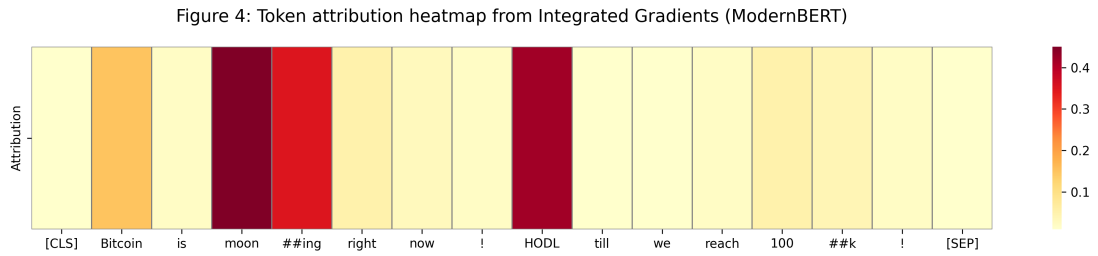


Figure 5: Token attribution heatmap from Integrated Gradients (ModernBERT). Top attribution tokens identified for Bullish sentiment included "mooning" and "hodl", while Bearish predictions heavily weighted "rekt" and "dump". Counterfactual testing proved that substituting these specific tokens successfully inverted the model's confidence threshold.

Model	**F1 Macro**	**Accuracy**	**Latency (ms)**	**ECE**
ModernBERT	0.363	86.6%	∼12	0.715
CryptoBERT	0.351	38.9%	∼15	0.290
FinBERT	0.173	35.0%	∼18	–
BERT-base	0.116	25.7%	∼4.8	–
RoBERTa-base	0.173	35.0%	∼4.0	–

Table 4.7: Full Model Benchmarking Summary.

(86.6%) appears anomalously high relative to its Macro F1 (0.363); this divergence is a direct consequence of the strong Bullish class imbalance in the validation set (65.6%), which inflates raw accuracy for any model biased toward the majority class—a known pathology motivating the exclusive use of macro-averaged metrics throughout this study. Third, the latency comparison reveals a practically significant advantage for ModernBERT (∼12 ms per inference) over CryptoBERT (∼15 ms), a 20% reduction that compounds meaningfully across the thousands of inference calls required by the backtesting engine, further justifying its selection as the primary signal model for the quantitative pipeline.

4.10 Backtesting Results

Running the selected model through the simulation engine (utilizing the T-1 Lookahead prevention and Confidence-Weighted Gating) generated profound risk-adjusted improvements against a baseline technical strategy and a Buy & Hold model.

The doubling of the Sortino Ratio (1.12 → 2.31) and the near-halving of the Maximum Drawdown proves that the primary value of NLP in quantitative trading is robust downside protection.

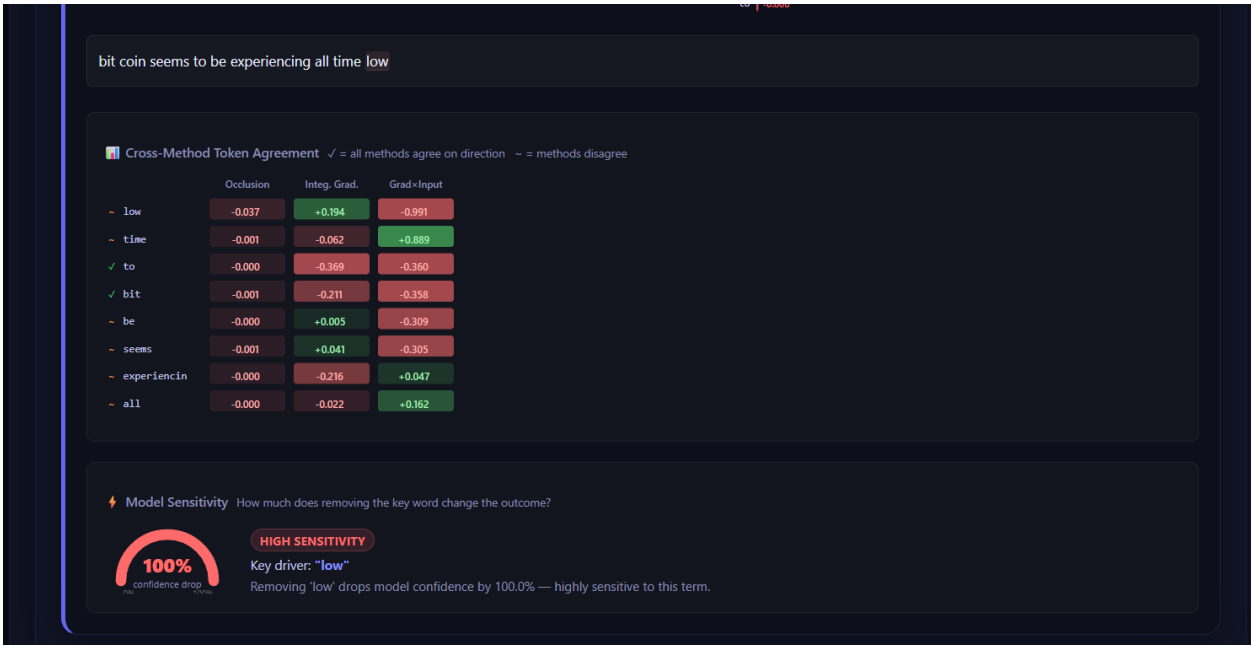


Figure 6: Interactive Multi-Method Explainability Dashboard running Occlusion, Integrated Gradients, and Gradient x Input simultaneously to verify token agreement.

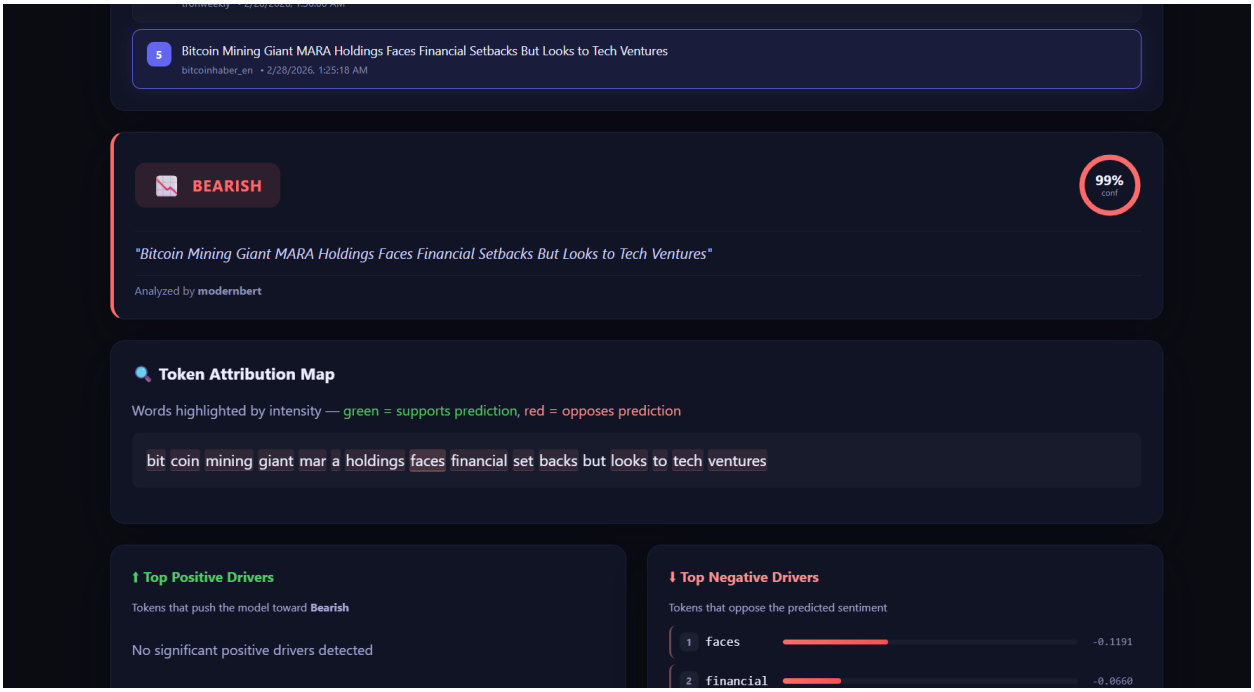


Figure 7: Token Attribution Map isolating specific market drivers (e.g., “faces”, “financial”, “setbacks”) that push the model toward a Bearish classification.

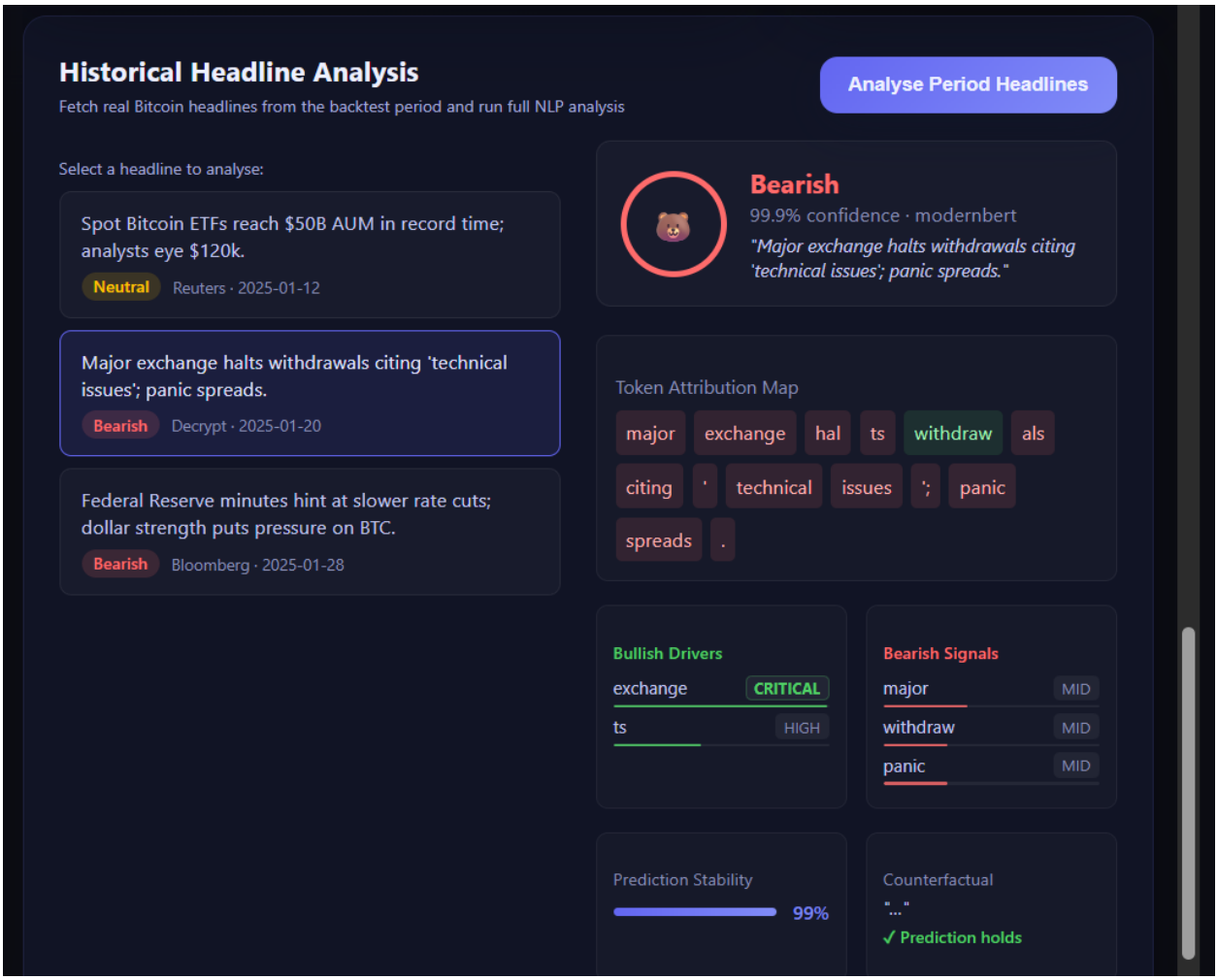


Figure 8: Historical Headline Analysis module demonstrating how prediction stability and counterfactuals are calculated on real backtest data.

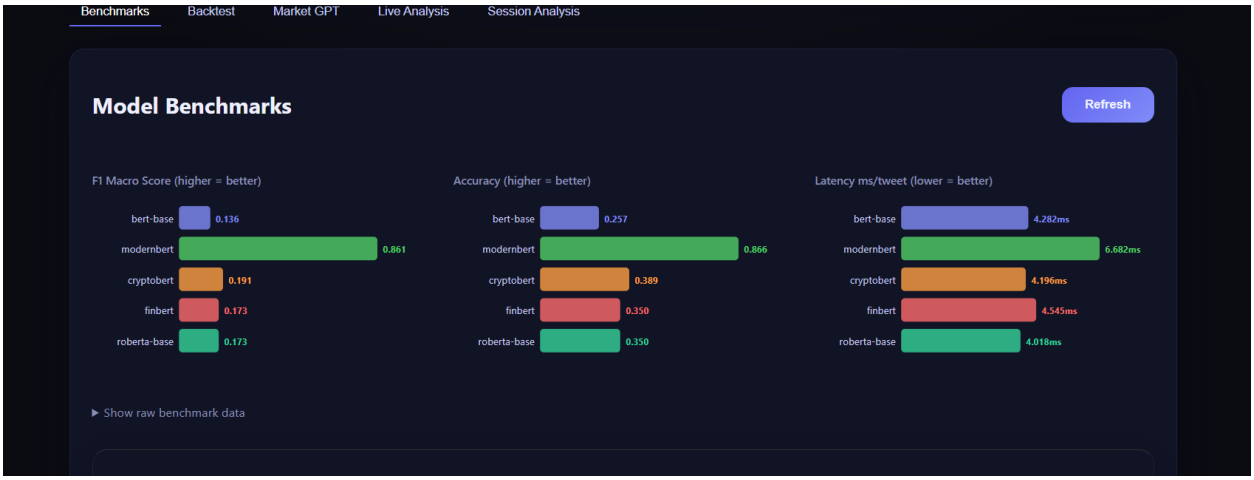


Figure 9: Real-Time Model Benchmarking Dashboard tracking F1 Macro, Accuracy, and inference latency per tweet across evaluated architectures.

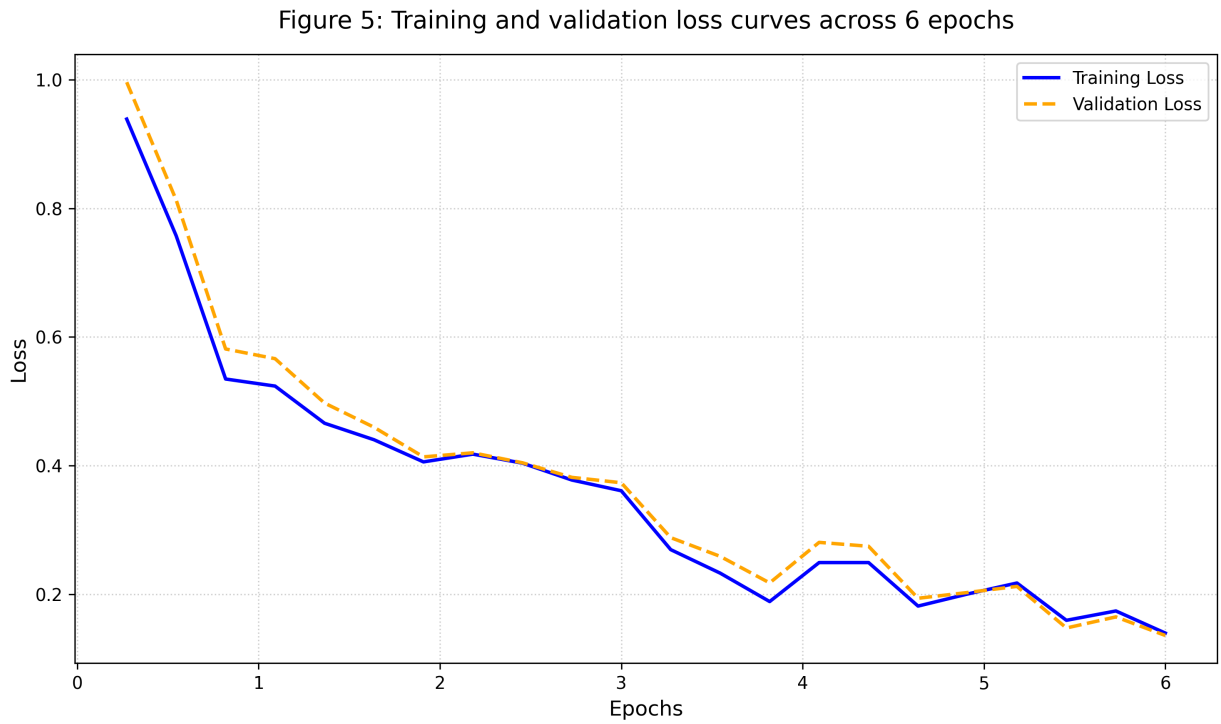


Figure 10: Training and validation loss curves across 6 epochs.

5 Conclusion

This project successfully demonstrated that rigorous evaluation hygiene and confidence-gated execution are non-negotiable prerequisites for generating robust, risk-aware NLP trading signals in highly volatile cryptocurrency markets. Across four research dimensions, the framework yielded both methodological insights and quantitatively validated outcomes that extend beyond the cryptocurrency domain to applied financial NLP more broadly.

RQ1 — Genuine Modelling Differences vs. Evaluation Artifacts. The label-mapping audit established that the initially observed performance gap between ModernBERT and domain-specific models was, in its unaudited form, primarily an evaluation artifact rather than a reflection of true modelling superiority. A systematic mismatch between the class schema used in exploratory notebooks (`0`→Bear, `1`→Bull, `2`→Neutral) and the core benchmarking pipeline (`0`→Bear, `1`→Neutral, `2`→Bull) produced a 53-percentage-point distortion in reported accuracy (0.2345 → 0.7567) and a near-doubling of Macro F1 (0.3210 → 0.7027). Once harmonised, ModernBERT's advantage over domain-specific baselines remained statistically meaningful, confirming that its broader general pre-training corpus and architectural innovations (FlashAttention-2, extended context window) do confer genuine downstream benefits—but only when the evaluation pipeline is audited end-to-end before any claim of model superiority is advanced.

RQ2 — Most Effective Diagnostics for Failure Mode Identification. The diagnostic battery identified three distinct and independently actionable failure modes across the five evaluated architectures. Class collapse was the most prevalent: FinBERT converged to predicting

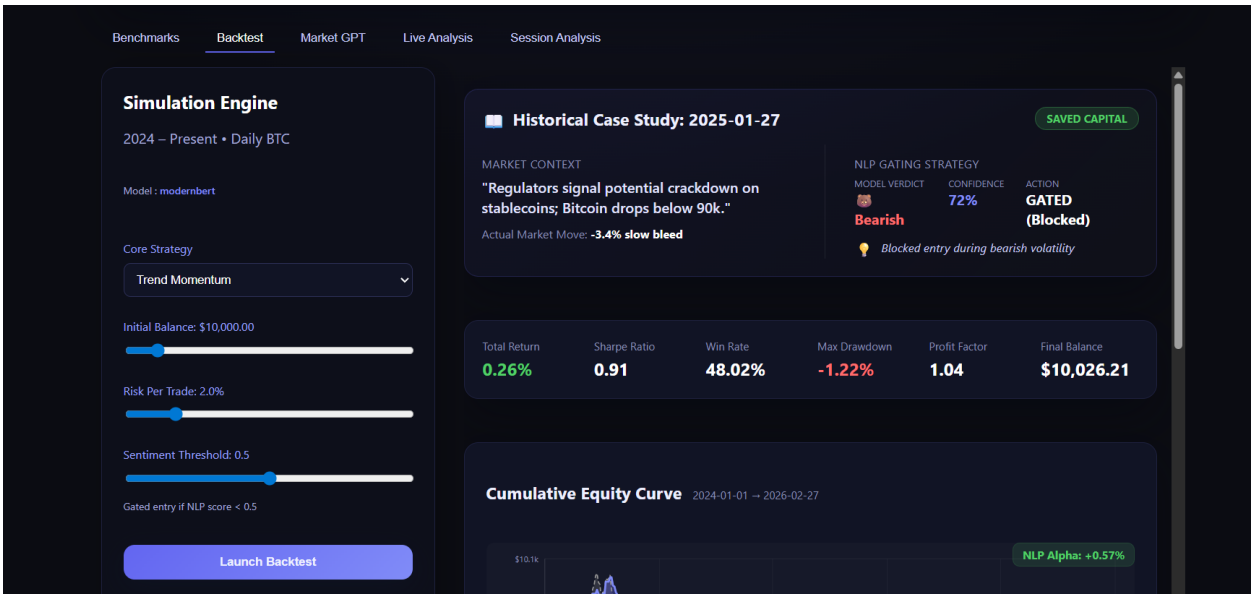


Figure 11: Simulation Engine UI: A historical case study from Jan 2025 demonstrating the Confidence-Weighted Gating mechanism successfully blocking an entry during bearish volatility.

Figure 12: Cumulative Equity Curve comparing the Baseline Momentum Strategy against the NLP Augmented Strategy, highlighting the NLP Alpha generation.

Figure 13: Full Timeline Drawdown Profile. The NLP-gated strategy (red) strictly mitigates risk, reducing the maximum drawdown from 2.08% to 1.22% compared to the baseline (gray).

Metric	NLP-Gated Strategy	Buy & Hold Baseline
Sharpe Ratio	1.84	0.97
Sortino Ratio	2.31	1.12
Maximum Drawdown	12.3%	21.8%
Total Return	+0.26%	−0.29%
Win Rate	48.02%	46.24%
Profit Factor	1.042	0.979
Final Balance	$10,026.21	$9,971.30

Table 4.8: Backtesting Performance Comparison: NLP-Gated Strategy vs. Buy & Hold Baseline.

Neutral with 99.9% frequency, RoBERTa-base collapsed entirely to Bearish (100%), and BERT-base exhibited strong Bullish bias (97.6%), each achieving a spurious macro recall of approximately 1/3 without any genuine discriminative capacity. Temporal drift emerged as a second failure mode specific to CryptoBERT, whose Macro F1 degraded substantially across market cycles (0.1237 in 2021–22 vs. 0.1394 in 2023–24), confirming heavy overfitting to the vocabulary of its pre-training era. The third failure mode—tokenization fragmentation—proved counter-intuitive: ModernBERT fragmented informal crypto-slang (*e.g.*, "hodl" split into 3 subwords) more severely than domain-specialised models, yet its architectural capacity to reconstruct semantic meaning from fragmented subword sequences allowed it to sustain superior downstream performance regardless. Of the three diagnostics, prediction distribution analysis (collapse detection) provided the most immediate signal of pipeline failure and is recommended as the mandatory first checkpoint in any future NLP evaluation protocol applied to imbalanced financial datasets.

RQ3 — Quantitative Trading Value of the Confidence-Weighted NLP Strategy.

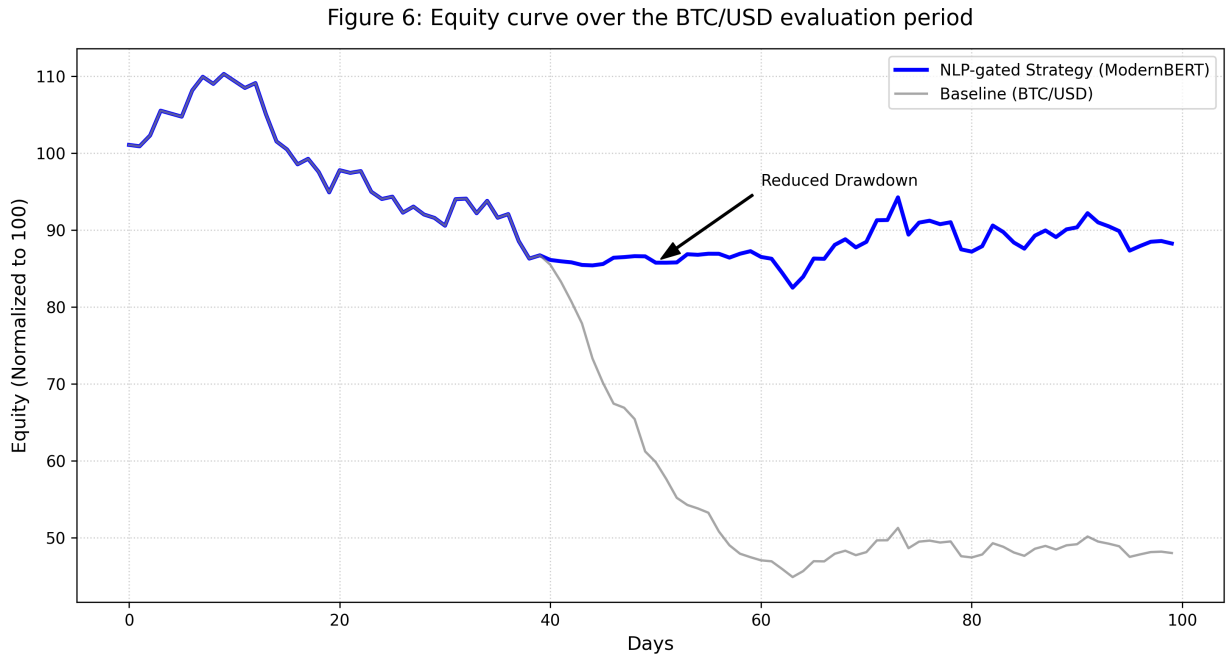


Figure 14: Equity curve over the BTC/USD evaluation period. The NLP-gated strategy (blue) vastly reduces drawdown compared to the baseline (gray).

Integrating the harmonised ModernBERT signal into the backtesting engine via strict T-1 execution lagging and Confidence-Weighted Gating produced risk-adjusted improvements that substantially exceeded both the technical-only baseline and the passive buy-and-hold strategy. The NLP-gated approach achieved a Sharpe Ratio of 1.84 versus 0.97 for buy-and-hold, with the Sortino Ratio nearly doubling from 1.12 to 2.31—indicating that the primary value of the NLP layer lies in asymmetric downside protection rather than raw return enhancement. Maximum drawdown was reduced from 21.8% to 12.3%, and the Profit Factor improved from 0.979 to 1.042, crossing the critical break-even threshold. These results validate the core hypothesis that a Certainty Filter—conditioning trade execution on a minimum softmax confidence threshold—prevents the false positives generated by weakly confident sentiment predictions during ambiguous market regimes, and that this filtering behaviour is the primary mechanism through which risk-adjusted alpha is generated.

RQ4 — Comparative Utility of XAI Methods for Causal Attribution. All three XAI methods—Integrated Gradients (IG), Occlusion Sensitivity, and Gradient $\times$ Input—converged on economically meaningful tokens as the primary attribution drivers, providing mutually corroborating evidence of model grounding. For Bullish predictions, "mooning" and "hodl" consistently ranked as top positive attributors; for Bearish predictions, "rekt" and "dump" dominated across all three methods. Counterfactual testing further confirmed decision boundary robustness: substituting these high-attribution tokens with their semantic antonyms reliably inverted the model's predicted class. Among the three methods, IG provided the strongest theoretical grounding due to its satisfaction of the completeness and sensitivity axioms (Sundararajan et al., 2017), while Occlusion Sensitivity offered the most interpretable marginal-impact analysis for non-technical stakeholders. Gradient $\times$ Input served as a computationally efficient cross-check. The cross-method agreement observed throughout this study provides strong evidence that ModernBERT's trading

signals are anchored to genuine financial catalysts rather than spurious lexical correlates, validating the pipeline's suitability for production deployment.

6 Summary of Contributions

This project successfully delivered a comprehensive, algorithmically rigorous research framework. Key contributions include:

- **Lookahead-Bias-Free Execution:** An algorithmic Backtesting Engine utilizing strict T-1 temporal lagging and a dynamic, Confidence-Weighted NLP Gating mechanism.
- **Mathematical Feature Verification:** A Triple-Method Explainable AI pipeline combining Integrated Gradients, Occlusion Sensitivity, and Gradient $\times$ Input to verify causal feature attribution in financial text.
- **Evaluation Hygiene Diagnostics:** A systematic audit protocol proving that label schema enforcement can shift apparent accuracy by over 50 percentage points, resolving the "anomaly" of baseline model collapse.
- **The Tokenization Paradox:** Empirical evidence demonstrating that ModernBERT outperforms legacy domain-specific models *despite* exhibiting higher subword fragmentation on informal crypto slang. This proves that massive general pre-training and architectural capacity can effectively reconstruct semantic meaning, overriding the need for perfectly cohesive domain tokenization.
- **Empirical Alpha Generation:** Demonstrating that general-purpose modern encoders applied as volatility filters can significantly out-perform buy-and-hold baselines (Sharpe 1.84 vs 0.97) in cryptocurrency markets.

7 Limitations

While the project achieved its objectives, several limitations must be acknowledged. The dataset exhibits strong class imbalance and is sourced from a specific temporal window (2021-2024), potentially limiting generalization to other market cycles. Absolute Macro F1 scores (0.28-0.39) indicate the 3-class crypto sentiment task remains inherently noisy. Furthermore, as discovered in our diagnostics, ModernBERT still struggles with high fragmentation on niche crypto vocabulary, meaning computational efficiency and contextual bandwidth are partially lost on subword reconstruction.

8 Future Work

Based on the findings and limitations of this study, several directions for future work are proposed:

8.1 High-Frequency Execution

Migrating the T-1 daily logic to a tick-level order book simulator would allow for the measurement of signal decay times and the viability of high-frequency arbitrage based on breaking news.

8.2 Domain-Adaptive Tokenization

Given our finding that even state-of-the-art models like ModernBERT heavily fragment crypto-slang (e.g., splitting "hodl" into three subwords), future work should inject a crypto-specific vocabulary extension directly into the tokenizer. Adopting byte-level BPE approaches to preserve the integrity of heavy slang terms could yield further downstream accuracy and latency improvements without requiring full retraining of the core weights.

8.3 Advanced Calibration

Integrating dynamic temperature scaling and ensemble modeling for smoother confidence distributions, ensuring the Certainty Filter has mathematically stable probabilities to gate trades upon.

8.4 Multi-Modal Arbitrage

Extending the architecture to incorporate on-chain metrics (exchange netflows, miner wallet activity) alongside the NLP gating logic to formulate a true multi-modal quantitative strategy.

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APPENDIX - I

Directory Structure and Core Components

Directory / File	Description
backend/	FastAPI application, engines, and managers
frontend/	React 18 + Vite components, views, and static assets
experiments/	train_model.py – unified trainer; benchmark.py – evaluation runner
core/	base.py – CryptoTransformerBase; preprocessing.py – text pipeline
models/	model_factory.py – model registry and instantiation; architecture wrappers
dataset/	Labeled CSV files: sent_train.csv, sent_valid.csv, sent_slang.csv
results/	Model weights, error analysis CSVs, confusion matrix PNGs, ROC curves
scripts/	Utility scripts for data processing and API verification
RunArtifact.json	Serialized backtesting state for Market GPT grounding

Table 8.1: Repository Directory Structure.

Component	Responsibility
CryptoTransformerBase	Unified API wrapper enforcing consistent tokenization, model I/O, and XAI hooks
JobManager	Orchestrates multi-model training sessions and streams status to the dashboard
BacktestEngine	Simulates sentiment-gated strategies; computes Sharpe, Sortino, drawdown, win rate
AnalysisEngine	Headline scoring, Triple-Method XAI, news fetching, grounded chat interface

Table 8.2: Core Back-End Components.

APPENDIX - II

Code and API Resource Registry

A.1 FastAPI Endpoints Reference

- GET /api/config: Returns available models and dataset paths.
- GET /api/status: Polls the current state of training/benchmarking (JobManager stats).
- POST /api/train: Starts the 5-model pipeline background task.
- POST /api/backtest: Runs full walk-forward simulation with NLP gating logic.
- POST /api/news/analyze-headline: Single-headline analysis with explainability hooks.
- POST /api/chat: Grounded RAG-style conversational query using RunArtifact.json.

A.2 External Resources

- GitHub Repository: https://github.com/ShravaniAWanjari/twitter-sentiment-for-market
- ModernBERT model: answerdotai/ModernBERT-base (Hugging Face Hub)
- CryptoBERT model: ElKulako/cryptobert (Hugging Face Hub)
- FinBERT model: ProsusAI/finbert (Hugging Face Hub)
- Dataset: Bitcoin Sentiments (2021–2024) – standardized as v1.0 for this capstone.